Submission CALC1-PS2-SUB01

Question CALC1-PS2-Q01
scratch: show step
1. $f'(x)=2x(6x-1)+(x^2+4)6$
2. $f'(2)=4(11)+(8)6=22$
=> FINAL: 22 (check units/interpretation)

Question CALC1-PS2-Q02
scratch: show step
1. $\int 2x\,dx = (2/2)x^2$
2. $[2/2\ x^2]_0^2 = 4$
=> FINAL: 4 (check units/interpretation)

Question CALC1-PS2-Q03
scratch: show step
1. $2w+2l=20 \Rightarrow l=10-w$
2. $A(w)=w(10-w)$; at $w=7$, $A=21$
3. A maximum requires $A'(w)=0$, so the stated dimensions are not assumed optimal without that check.
=> FINAL: $A(w)=w(10-w)$; $A(7)=21$; check $A'(w)=0$ (check units/interpretation)

Question CALC1-PS2-Q04
scratch: show step
1. $x^2-9=(x-3)(x+3)$
2. cancel $x-a$ only after identifying $x\neq a$
3. $\lim_{x\to 3} x+3=6$
=> FINAL: 6 (check units/interpretation)

Question CALC1-PS2-Q05
scratch: show step
1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from + to – indicates a local maximum.
=> FINAL: critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum (check units/interpretation)

Question CALC1-PS2-Q06
scratch: show step
1. $f'(x)=2x(1x-1)+(x^2+3)1$
2. $f'(3)=6(2)+(12)1=19$
=> FINAL: 19 (check units/interpretation)

Question CALC1-PS2-Q07
scratch: show step
1. $\int 4x\,dx = (4/2)x^2$

2. [4/2 x^2]_0^5 = 50
=> FINAL: 50 (check units/interpretation)

Question CALC1-PS2-Q08
scratch: show step
1. 2w+2l=16 ⇒ l=8-w
2. A(w)=w(8-w); at w=5, A=15
3. A maximum requires A'(w)=0, so the stated dimensions are not assumed optimal without that check.
=> FINAL: A(w)=w(8-w); A(5)=15; check A'(w)=0 (check units/interpretation)